

Manufacturing Economic Monitor: Relative Change Since 2017

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1 Question

Which supported manufacturing indicator moved farthest from its January 2017 baseline by the latest available observation?

2 Data and method

This report uses preserved monthly observations from the Federal Reserve Economic Data (FRED) series-observations API. The committed SQLite database is `data/manufacturing_monitor.db`. The selected series are `IPMAN`, `IPDMAN`, `IPNMAN`, `MCUMFN`, and `PCUOMFGOMFG`.

The analysis starts on **2017-01-01** and continues through the latest observation available in the preserved data. Original units differ across indicators, so the comparison uses the project's existing normalization method: the first valid observation for each series is set to 100. The evidence below is generated by `reports/render_report.py`, which calls `manufacturing_monitor.analytics.process_multiple_series` rather than recreating loading, validation, or analysis logic.

3 Generated answer

Answer: `PCUOMFGOMFG` moved farthest from its January 2017 baseline. Its latest normalized index is **148.55**, a change of **+48.55%** from the baseline value of 100.

Series	Latest date	Latest	raw value	Units	Normalized index	Change from baseline
IPDMAN	2026-06-01		101.6971	Index 2017=100, Season- ally Ad- justed	102.58	+2.58%
IPMAN	2026-06-01		98.6995	Index 2017=100, Season- ally Ad- justed	99.15	-0.85%
IPNMAN	2026-06-01		95.5366	Index 2017=100, Season- ally Ad- justed	95.54	-4.46%
MCUMFN	2026-06-01		75.5559	Percent, Season- ally Ad- justed	98.91	-1.09%
PCUOMFGOMFG	2026-06-01		275.702	Index Dec 1984=100, Not Sea- sonally Adjusted	148.55	+48.55%

The normalized measure answers a relative-change question without mixing the original units. The raw values remain available in the table for context, but direct comparisons across incompatible units should be avoided.

4 Figure

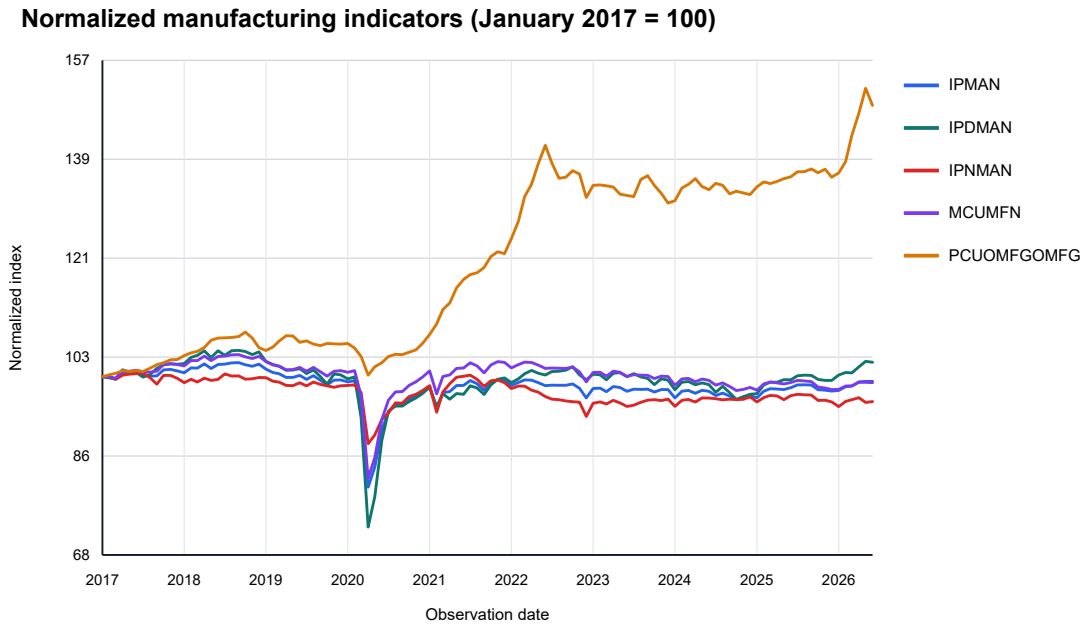


Figure 1: Normalized comparison of five supported manufacturing indicators.

5 Interpretation

The figure shows relative movement rather than direct original-unit comparison. A line above 100 indicates growth relative to the first valid observation in the selected period, while a line below 100 indicates decline. The largest distance from 100 identifies the indicator that changed most relative to its own baseline; it does not imply that its original-unit value is directly comparable with the other series.

6 Reproducibility

Restore the locked project environment and rebuild the report from the repository root with:

```
uv sync --dev
uv run python reports/render_report.py
quarto render reports/manufacturing_report.qmd --to typst
```

The report does not contact FRED and does not require an API key. It uses committed data and reusable project code. The rendered PDF is `reports/manufacturing_report.pdf`.